Nature-Inspired Algorithms

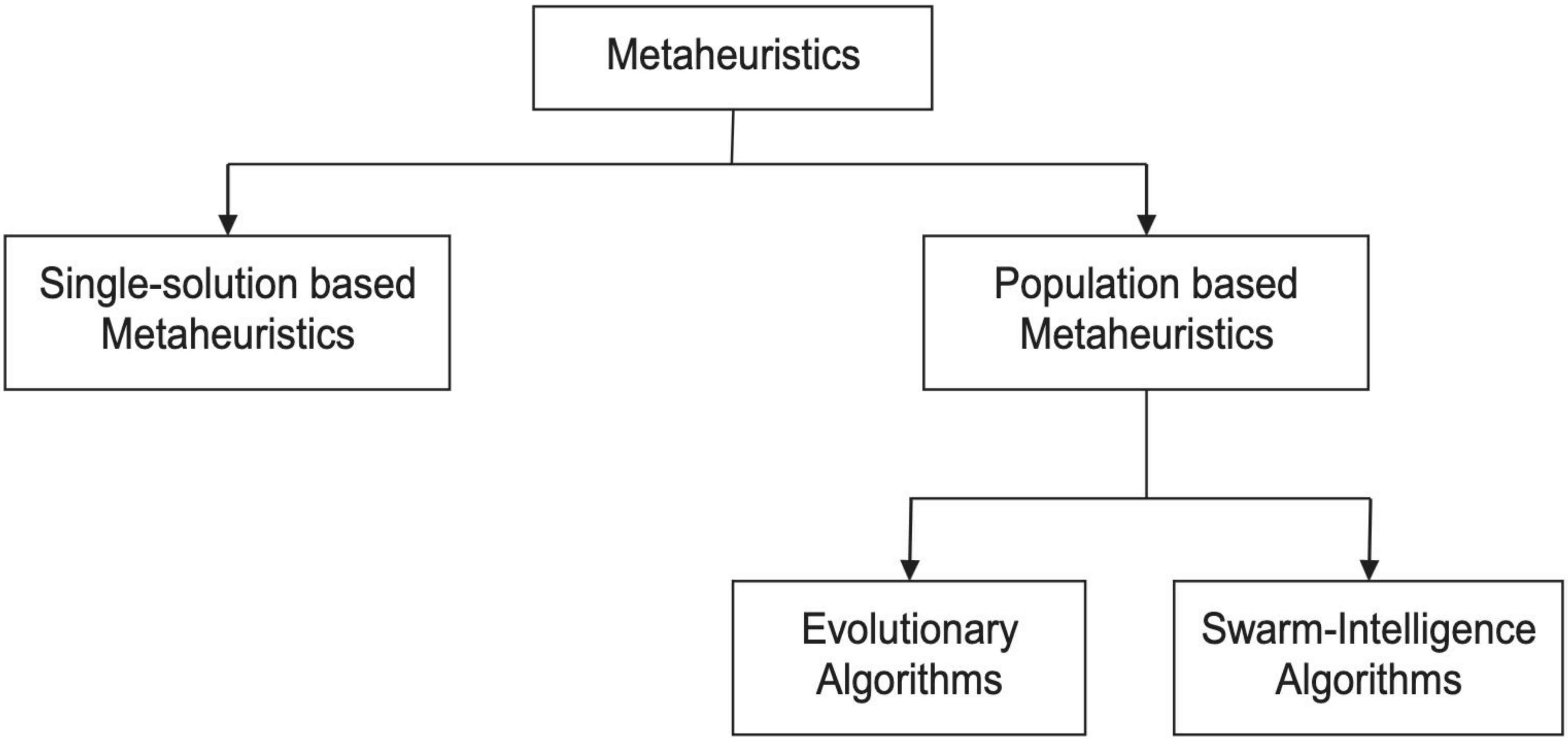
ENEL445 Applied Engineering Optimisation

2025

Outline

- Genetic Algorithm (GA)
- Particle Swarm Optimisation (PSO)

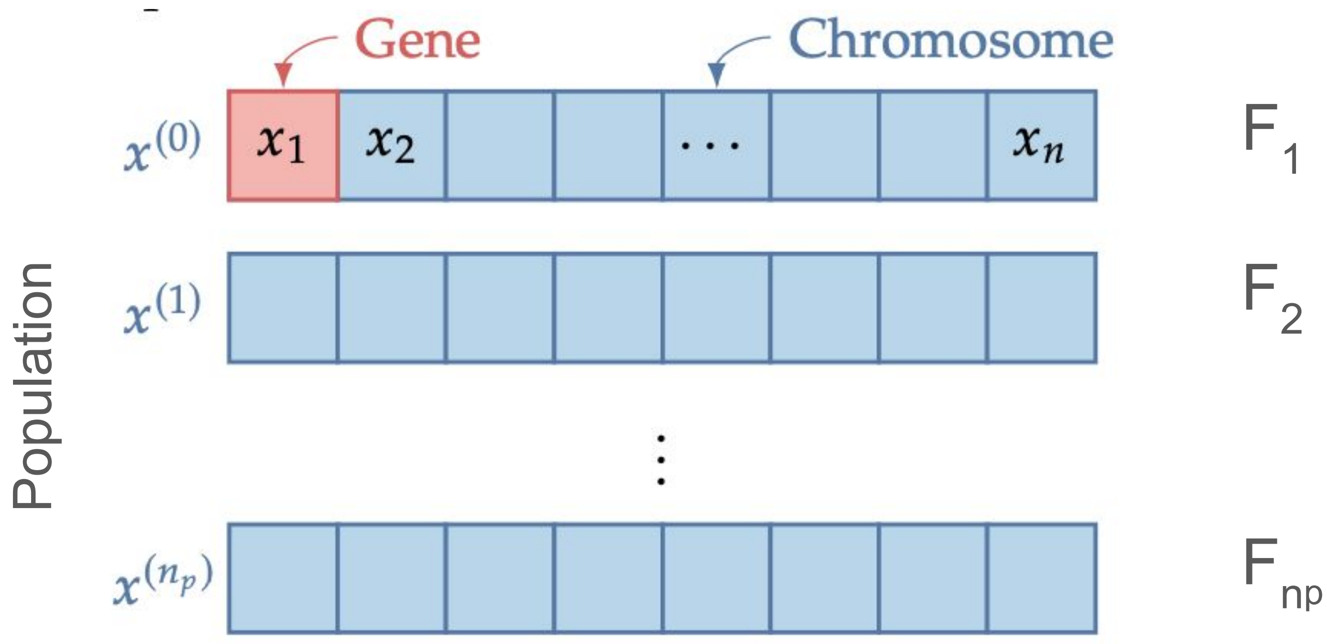
Nature-inspired Algorithms



A higher-level strategy **(meta-)** that guides or controls **heuristic** methods to solve optimisation problems more effectively.

Genetic Algorithm - Terminology

- **Gene** A design variable
- **Chromosome** A design point / vector
- **Population** A set of design points / chromosomes
- **Fitness** Objective function value for a chromosome



Genetic Algorithm - Terminology

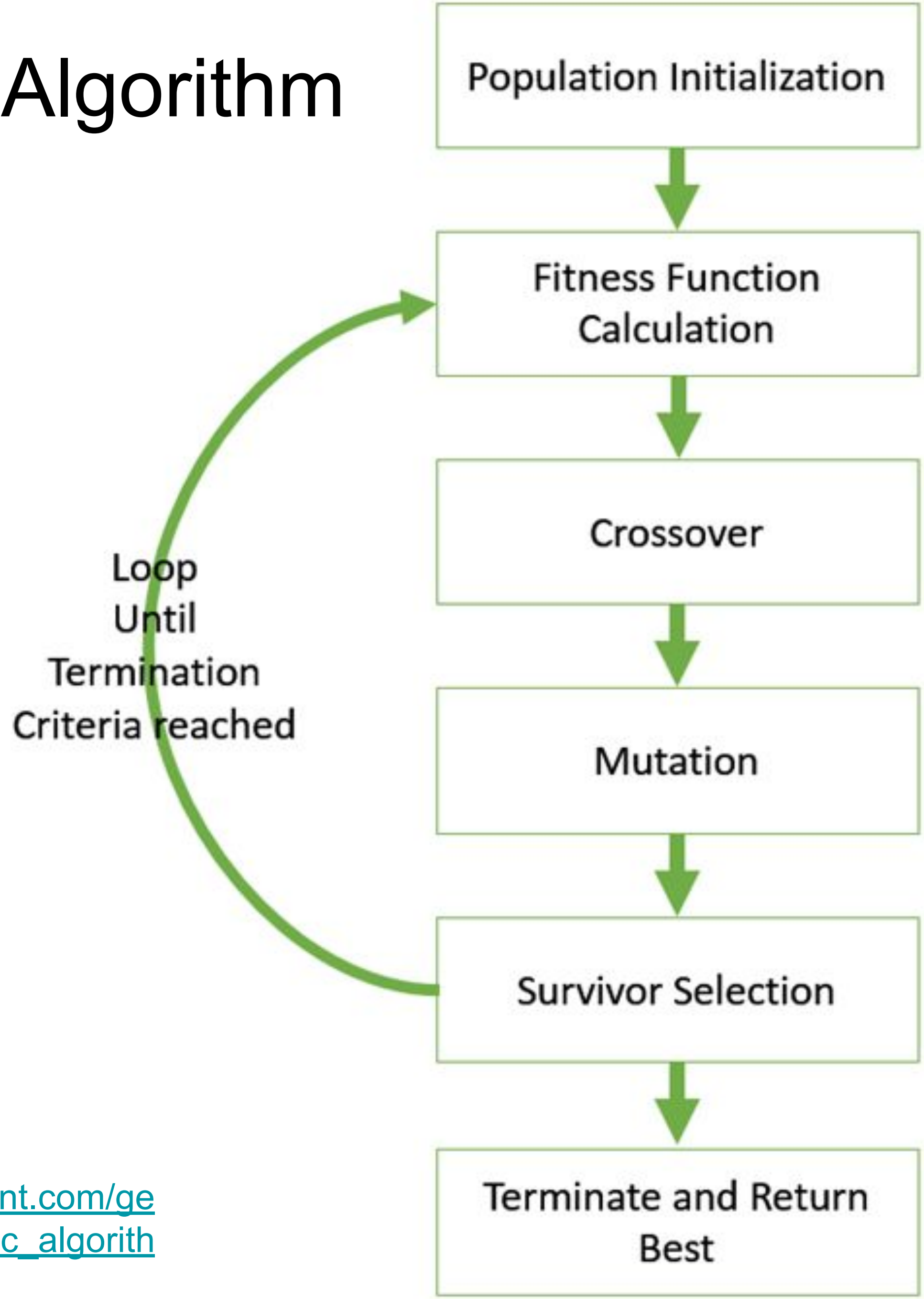
- **Generation** — One genetic algorithm iteration
- **Allele** — The value a gene takes for a particular chromosome
- **Gene Representation** — Method of encoding the numerical value of the design variable

Gene Representation

- **Binary-encoded** algorithms use bits to represent the design variables (also called bit-set or bit-string)
    - Each gene is represented as a string of 0 or 1

- **Real-encoded** algorithms use real-valued numbers to represent the design variables
    - Each gene is represented as a real number

Genetic Algorithm



https://www.tutorialspoint.com/genetic_algorithms/genetic_algorithms_fundamentals.htm

Genetic Algorithm

- GA starts with a set of design points (the population) rather than a single starting point, and each iteration (generation) updates this set in some way.

- GAs evolve the population using three main steps:

(1) **Selection** - Based on natural selection, where members of the population that acquire favourable adaptations are more likely to survive longer and contribute more to the population gene pool.

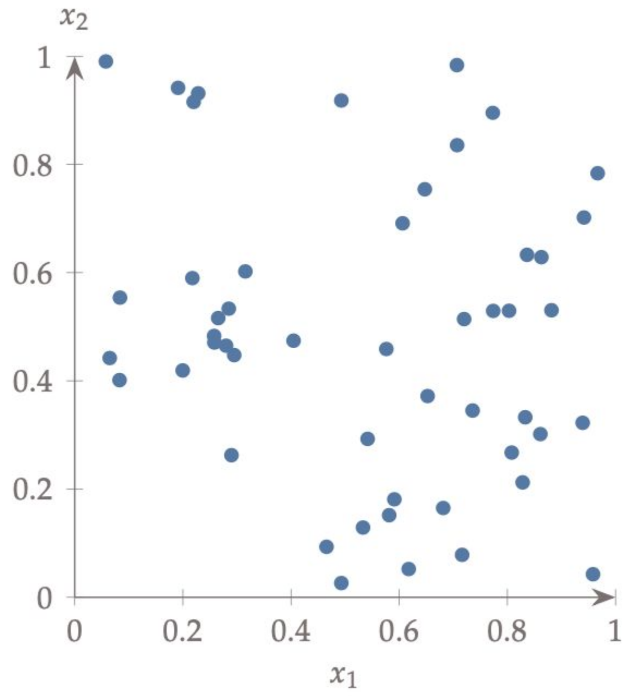
(2) **Crossover** - Inspired by chromosomal crossover, which is the exchange of genetic material between chromosomes during sexual reproduction.

(3) **Mutation** - Mimics genetic mutation, which is a permanent change in the gene sequence that occurs naturally.

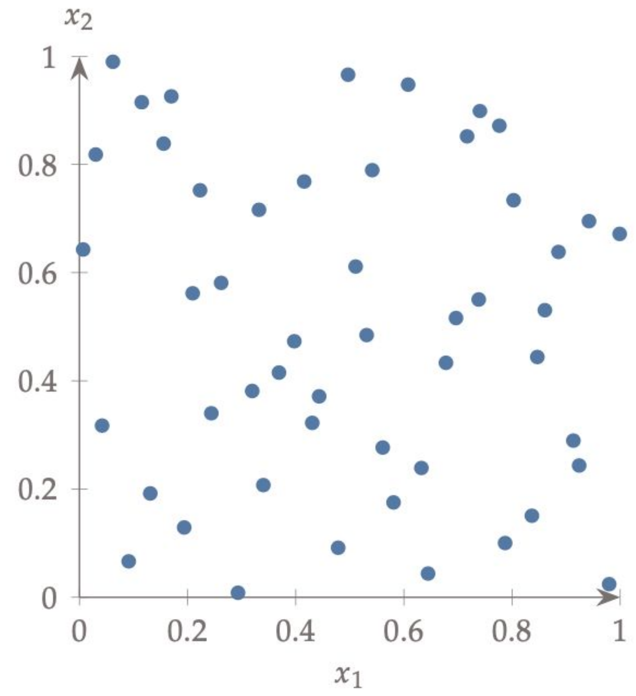
Initial Population

- The first step in a genetic algorithm is to generate an initial set (population) of points.

- As a rule of thumb, the population size should be approximately one order of magnitude larger than the number of design variables, and this size should be tuned.



Random



Latin hypercube sampling

Selection

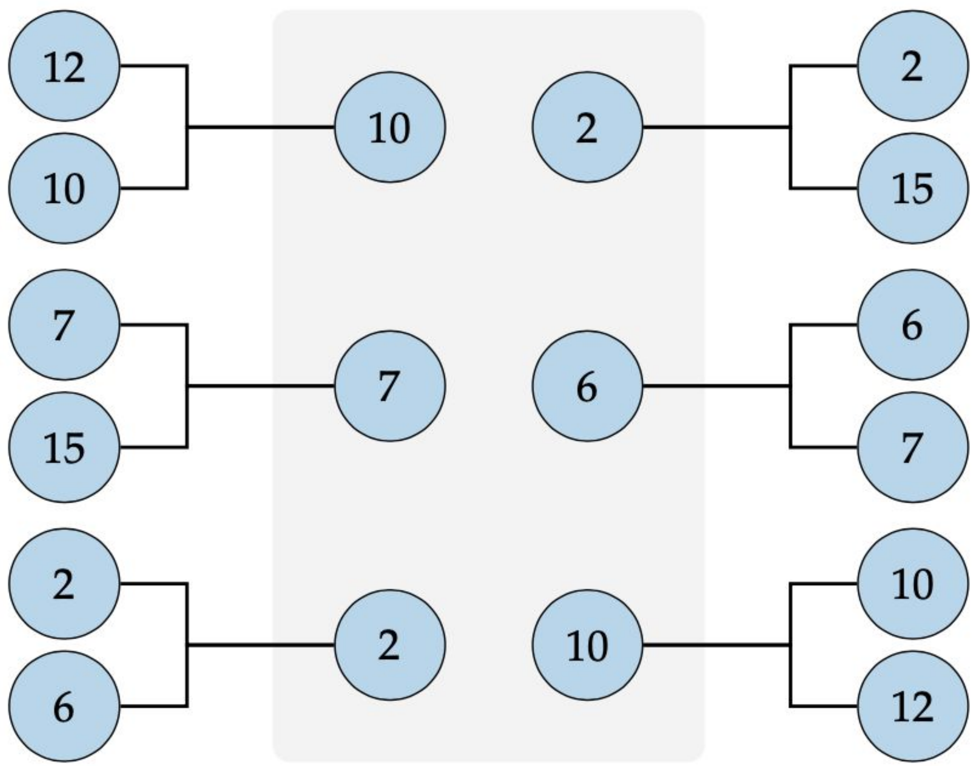
- We choose chromosomes from the population for reproduction/recombination (called a **mating pool**) - choosing parents!

- Desirable to choose a mating pool that **improves in fitness** (thus mimicking the concept of natural selection), but take care to also **maintain diversity**

- Two popular methods of selection are:

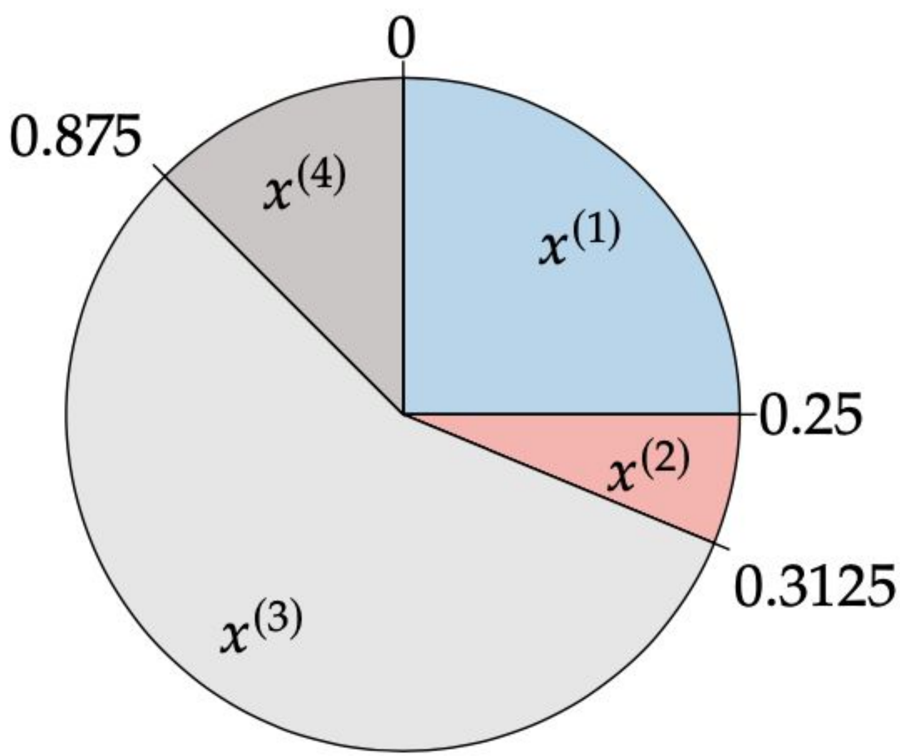
Selection

- **Tournament Selection**
  - Randomly select two chromosomes from the population until the requisite number of pairs is reached.
  - Compare the two chromosomes and select the one with the highest fitness to be in the mating pool



Selection

- **Roulette Wheel Selection**
  - Better points are allocated a larger sector on the roulette wheel to have a higher probability of being selected.
  - Voltage divider!



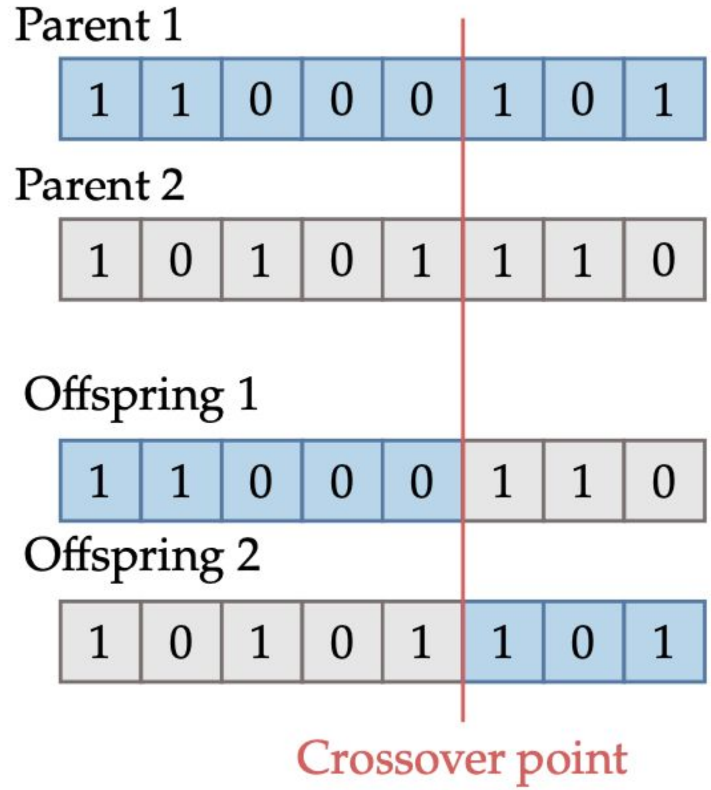
Crossover - Binary encoded

- Chromosomes in the mating pool are combined to generate new chromosomes

- **Single-point crossover** involves generating a random integer $1 \leq k \leq m-1$ that defines the crossover point

- For one of the offspring, the first k bits are taken from parent 1 and the remaining bits from parent 2, and vice versa

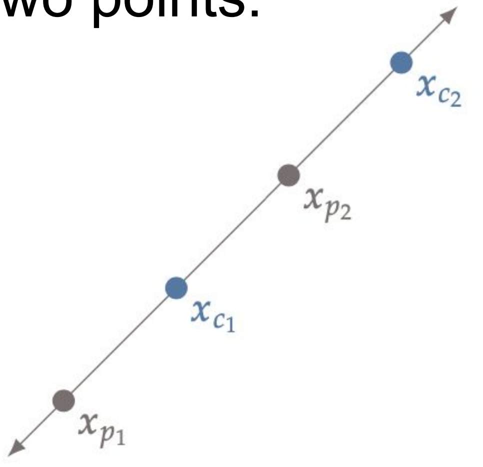
- Various other strategies are possible



Crossover - Real encoded

- There are various options for the reproduction of real-encoded chromosomes. A standard method is **linear crossover**, which generates two or more points in the line defined by the two parent points. One option for linear crossover is to generate the following two points:

$$x_{c_1} = 0.5x_{p_1} + 0.5x_{p_2}$$
$$x_{c_2} = 2x_{p_2} - x_{p_1},$$



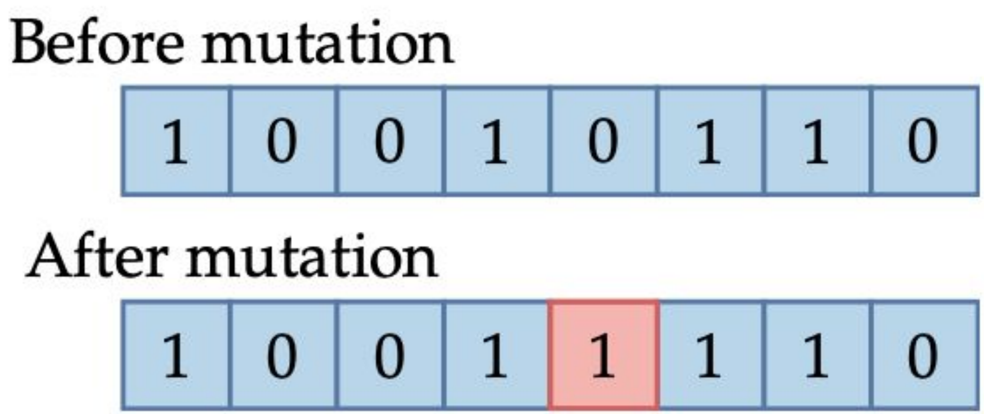
- Another option is a simple crossover like the binary case where a random integer is generated to split the chromosomes. This simple crossover does not generate as much diversity as the binary case and relies more heavily on an effective mutation.

Mutation

- Mutation is a **random operation** performed to change the genetic information and introduces additional diversity into the population.

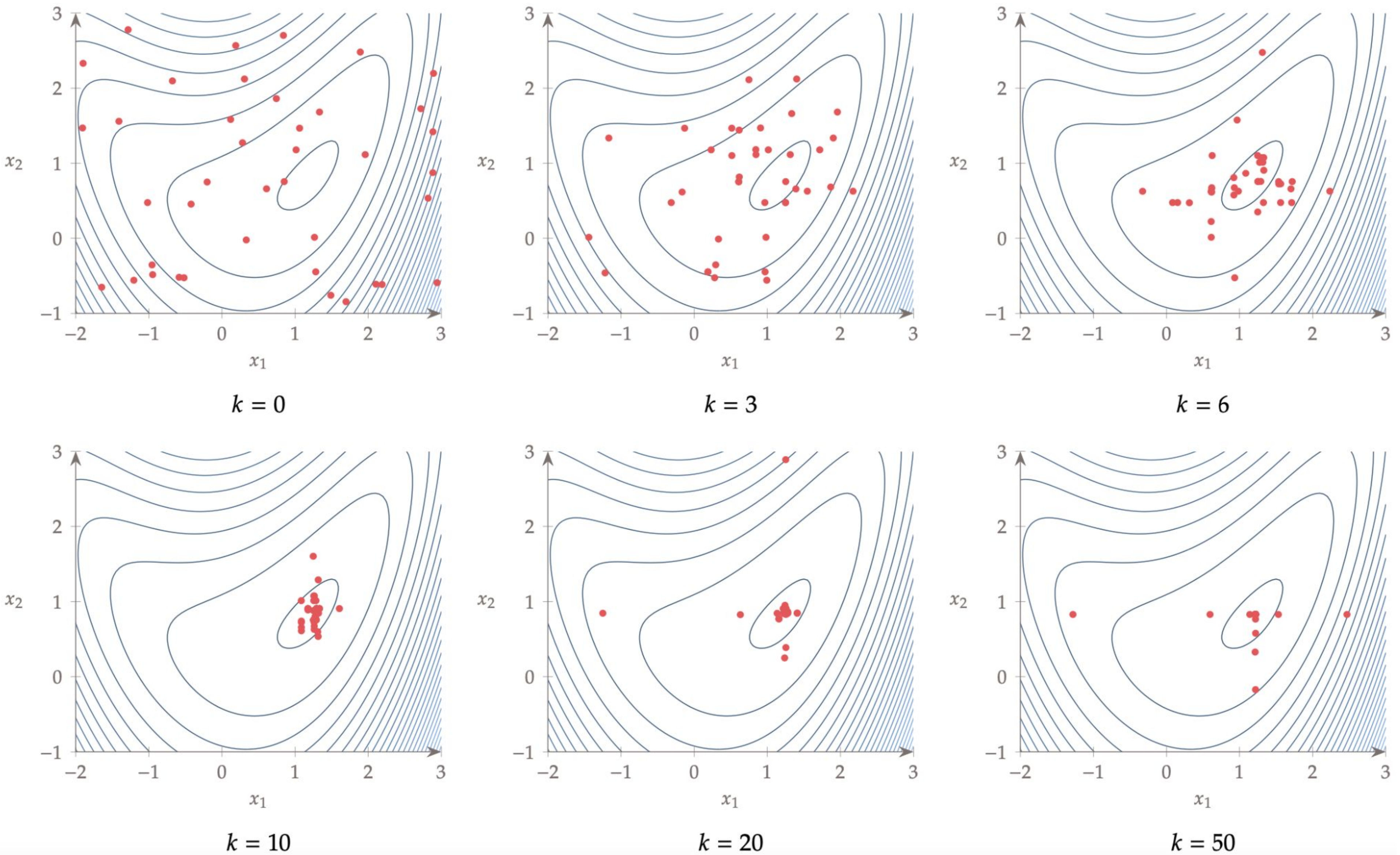
- Even though selection and reproduction effectively recombine existing information, we might need new information, or recover useful genetic information that may have been lost.

- Mutation for real-encoded chromosome is just adding a random number to each gene.



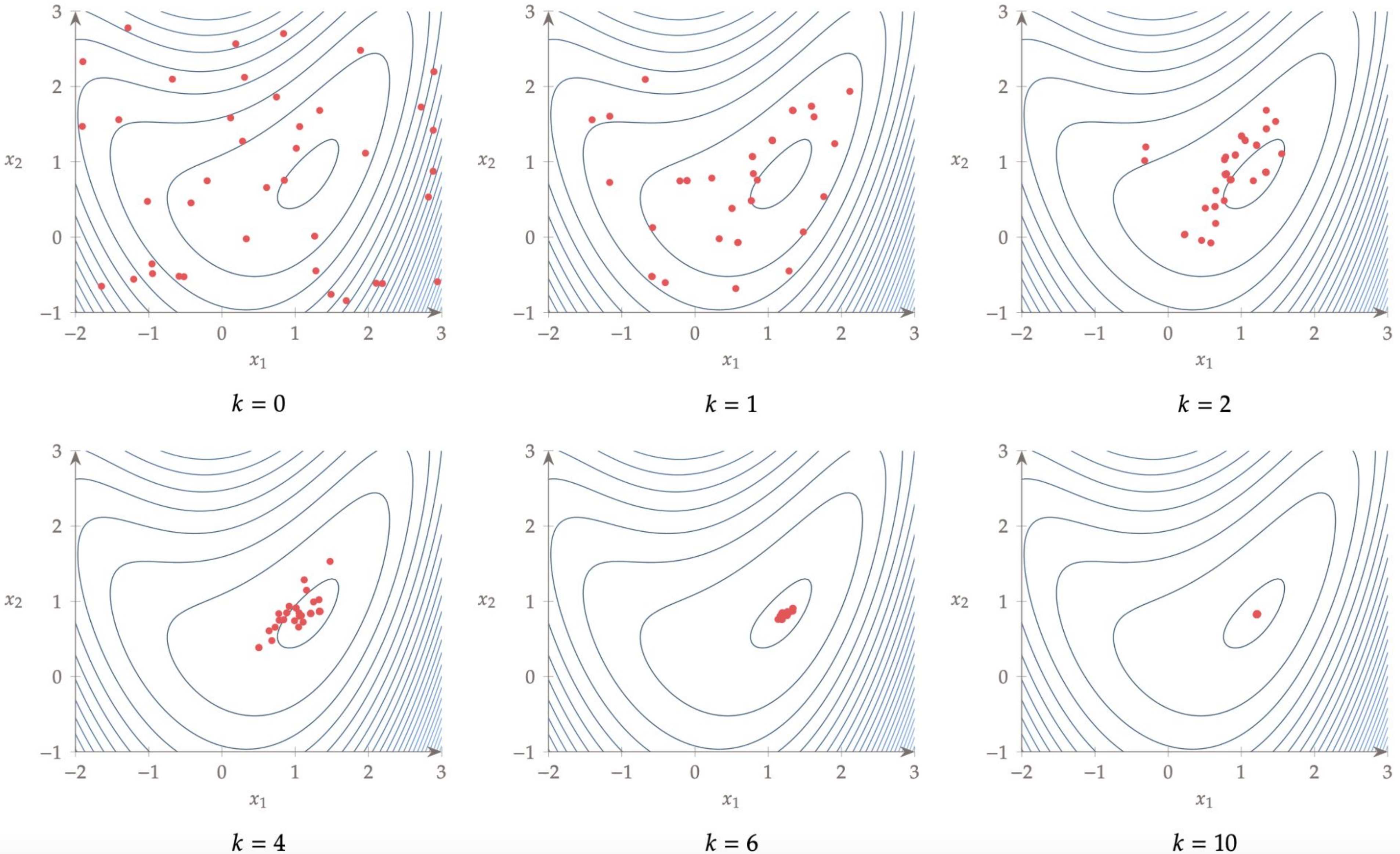
Genetic Algorithm - Results

Binary-encoded



Genetic Algorithm - Results

Real-encoded



Genetic Algorithm - Constraint Handling

- Constraints can be handled in many ways in a GA

- For example in tournament selection, we can choose among two competitors as follows:

    - Reject chromosomes that are infeasible.

    - Among two infeasible chromosomes, choose the one with a smaller constraint violation.

Genetic Algorithm - Convergence

- Rigorous mathematical convergence criteria, like those used in gradient-based optimisation, do not apply to GAs.

- The most common way to terminate a GA is to specify a maximum number of iterations, which corresponds to a computational budget.

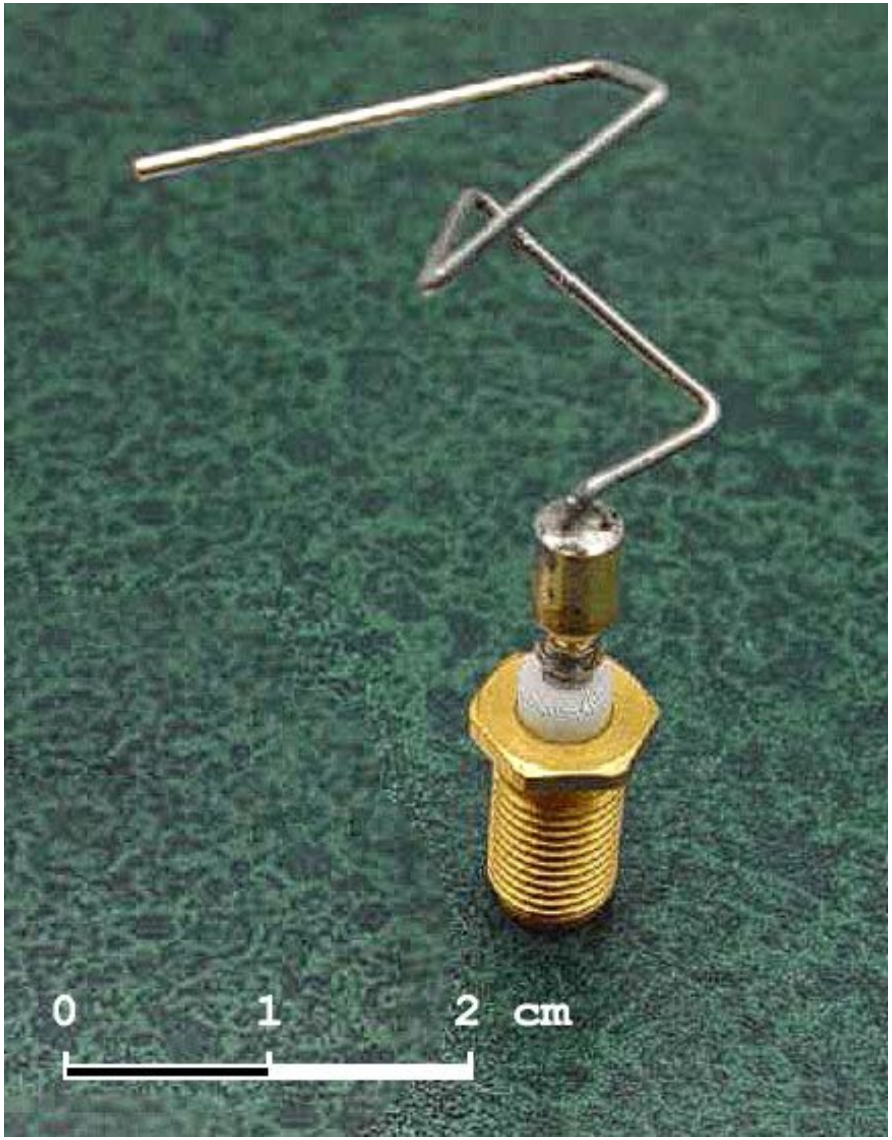
- Monitoring the trends in population fitness:
    - Average of the population's fitness
    - The fitness of the best member in the population

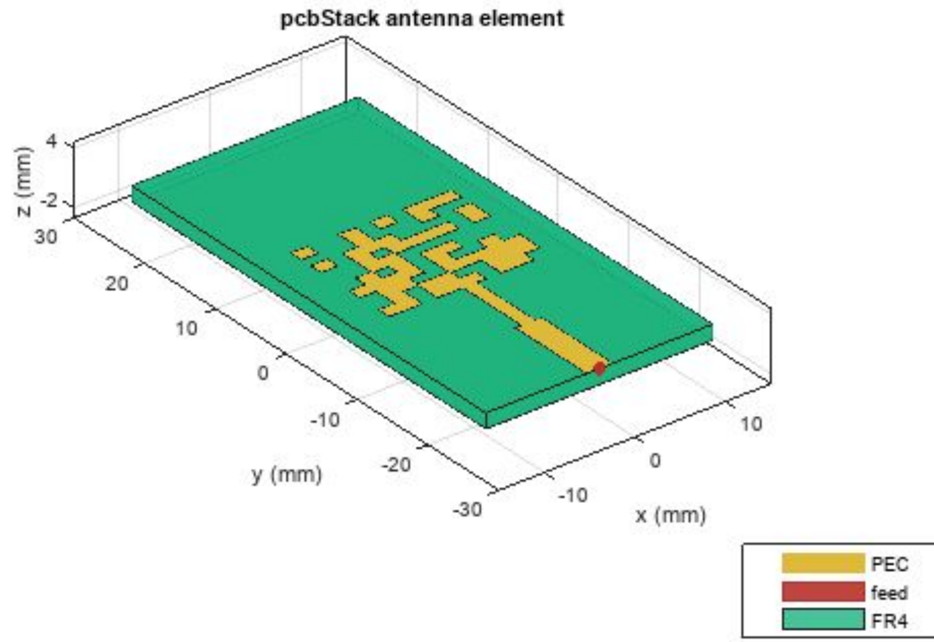
- The best member can disappear as a result of crossover or mutation. To avoid this and to improve convergence, many GAs employ **elitism**. This means that the fittest population member is retained to guarantee that the population does not regress.

Comparing Gene Representation

- **Binary-encoded** algorithms use bits to represent the design variables (also called bit-set or bit-string)
  - Each gene is represented as a string of 1 or 0
  - Provides **faster implementation** of crossover and mutation operators.
  - However, it requires **extra effort** to convert into binary form and **accuracy** of algorithm depends upon the binary conversion.

- **Real-encoded** algorithms use real-valued numbers to represent the design variables
  - Each gene is represented as a real number
  - Length of chromosome is **shorter**
  - Avoids the “**Hamming cliff**” issue of binary encoding, which is caused by the fact that many bits must change to move between adjacent real numbers (e.g., 0111 to 1000).





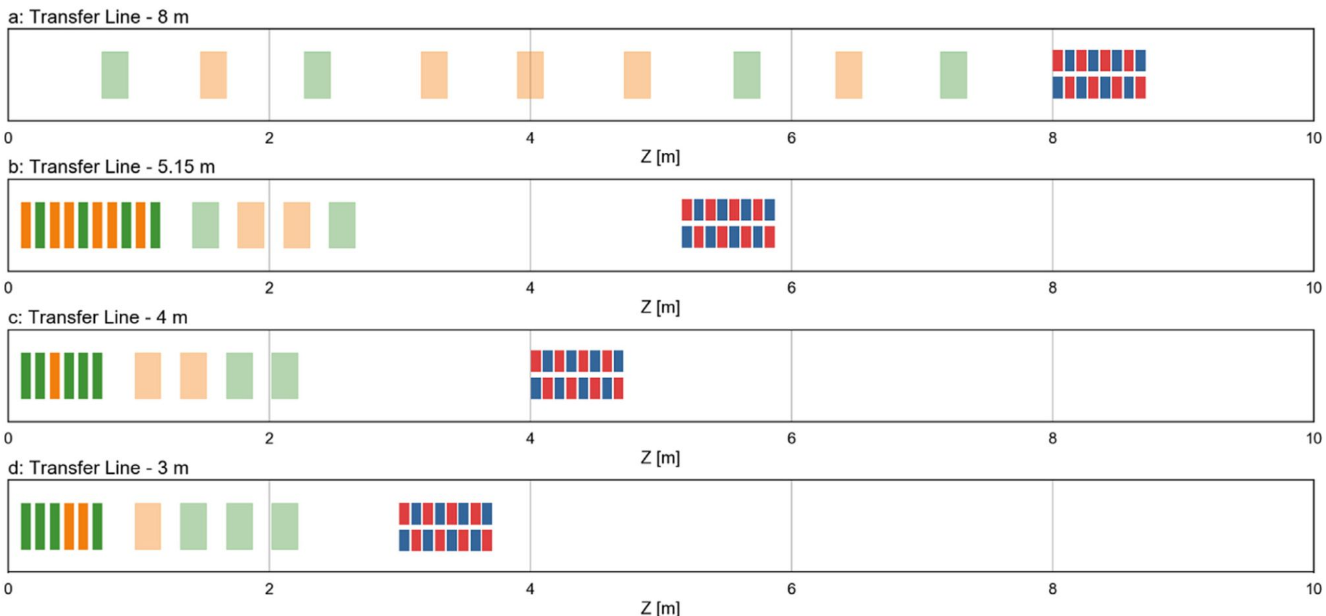
https://en.wikipedia.org/wiki/Evolved_antenna

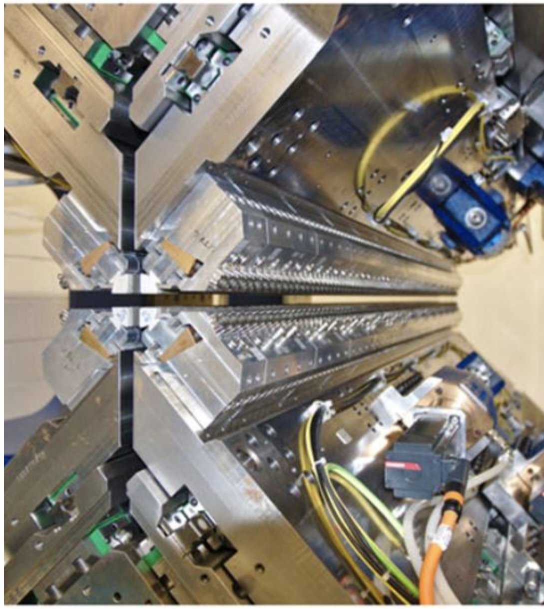
https://www.mathworks.com/help/antenna/ug/miniaturize-rectangular-microstrip-patch-antenna-using-genetic-algorithm.html

Electron beam transfer line design for plasma driven Free Electron Lasers

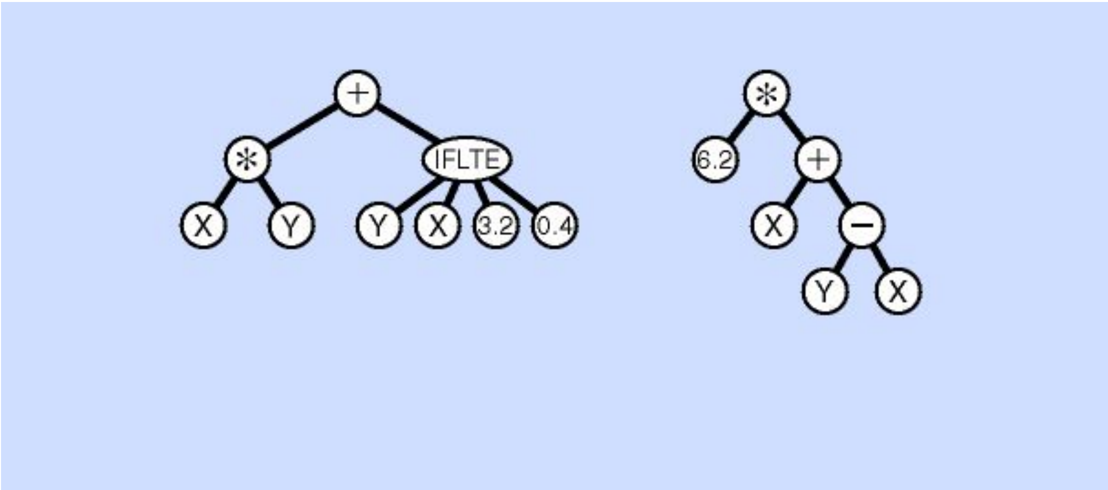
M. Rossetti Conti [a b], A. Bacci [a], A. Giribono [c], V. Petrillo [b], A.R. Rossi [a], L. Serafini [a], C. Vaccarezza [c]

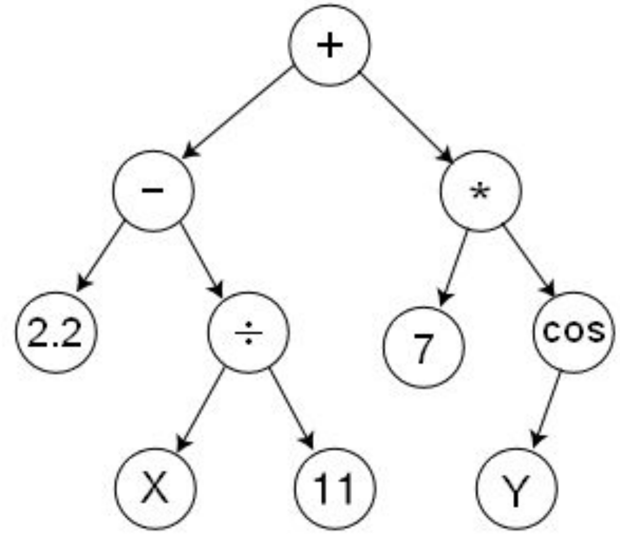
Plasma driven particle accelerators represent the future of compact accelerating machines and Free Electron Lasers are going to benefit from these new technologies. One of the main issue of this new approach to FEL machines is the design of the transfer line needed to match of the electron-beam with the magnetic undulators. Despite the reduction of the chromaticity of plasma beams is one of the main goals, the target of this line is to be effective even in cases of beams with a considerable value of chromaticity. The method here explained is based on the code GIOTTO (Bacci etal., 2016) that works using a homemade genetic algorithm and that is capable of finding optimal matching line layouts directly using a full 3D tracking code.



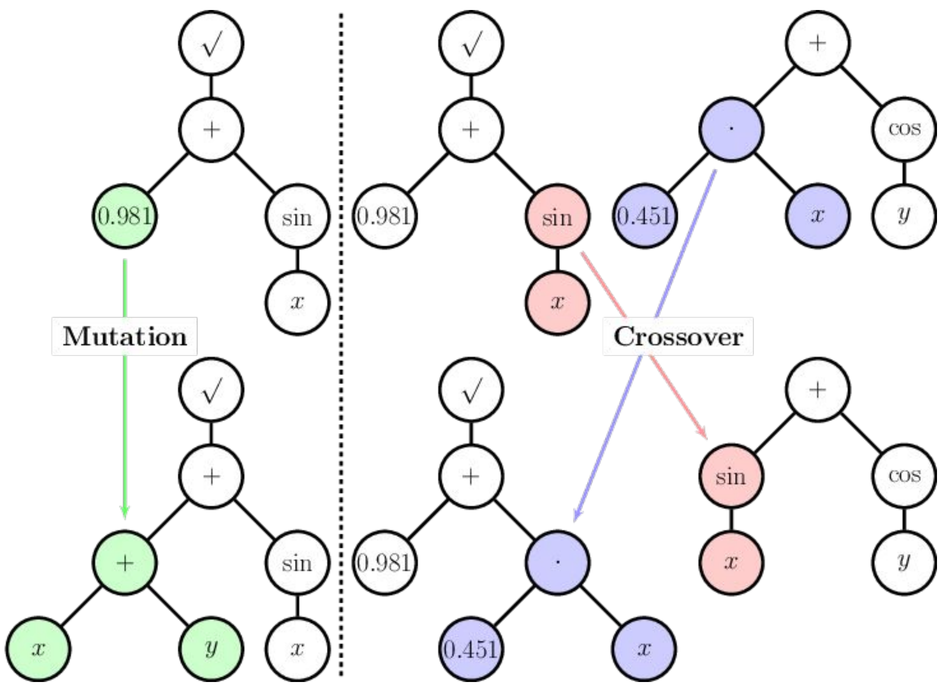


Genetic Programming





$$\left(2.2-\left(\frac{X}{11}\right)\right)+\left(7*\cos(Y)\right)$$



https://en.wikipedia.org/wiki/Genetic_programming

List of genetic algorithm applications

From Wikipedia, the free encyclopedia

Contents hide

This is a list of **genetic algorithm (GA) applications**.

Natural Sciences, Mathematics and Computer Science [edit]

- Bayesian inference links to particle methods in Bayesian statistics and hidden Markov chain mode
- Artificial creativity
- Chemical kinetics (gas and solid phases)
- Calculation of bound states and local-density approximations
- Code-breaking, using the GA to search large solution spaces of ciphers for the one correct decry
- Computer architecture: using GA to find out weak links in approximate computing such as lookah
- Configuration applications, particularly physics applications of optimal molecule configurations for
- Construction of facial composites of suspects by eyewitnesses in forensic science.[4]
- Data Center/Server Farm.[5]

https://en.wikipedia.org/wiki/List_of_genetic_algorithm_applications

Scipy implementation

Particle Swarm Optimisation (PSO)

PSO

Eberhart and Kennedy, “New optimizer using particle swarm theory”, 1995.

- Stochastic population-based optimisation algorithm like the GA.

- Inspired by “swarm intelligence” - property of a system whereby the collective behaviors of agents interacting locally with their environment cause coherent global patterns.

“Dumb agents, properly connected into a swarm, yield smart results”

PSO - Terminology

- **Particle / Agent** A design point / vector
- **Swarm** A set of design points

- The “swarm” in PSO is a set of design points that “move” in n-dimensional space, looking for the best solution.
- The history of each point is relevant to the PSO algorithm, so we adopt the term **particle**. Each particle moves according to a ***velocity***.
- The velocity of a particle changes according to the past objective function values of that particle and the current objective values of the rest of the particles.

PSO - Velocity

- Each particle remembers the location where it found its best result so far, and it exchanges information with the swarm about the location where the swarm has found the best result so far.

- The position of particle i for iteration k + 1 is updated according to:

$$x_{k+1}^{(i)} = x_k^{(i)} + v_{k+1}^{(i)} \Delta t$$

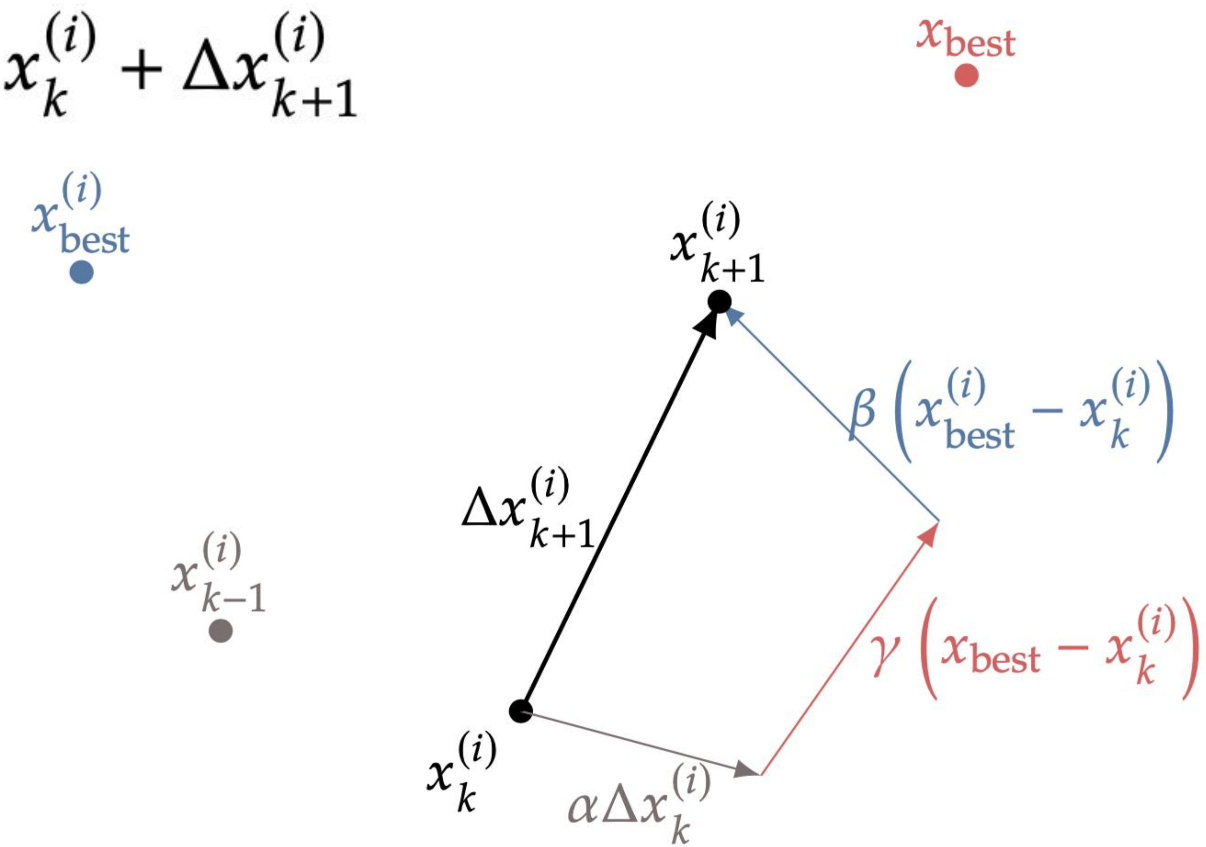
$$v_{k+1}^{(i)} = \alpha v_k^{(i)} + \beta \frac{x_{\text{best}}^{(i)} - x_k^{(i)}}{\Delta t} + \gamma \frac{x_{\text{best}} - x_k^{(i)}}{\Delta t}$$

PSO - Position update

- Because the time step is artificial, we can eliminate it by multiplying by Δt to yield the position update rule

$$\Delta x_{k+1}^{(i)} = \alpha \Delta x_k^{(i)} + \beta \left(x_{\text{best}}^{(i)} - x_k^{(i)} \right) + \gamma \left(x_{\text{best}} - x_k^{(i)} \right)$$

$$x_{k+1}^{(i)} = x_k^{(i)} + \Delta x_{k+1}^{(i)}$$



PSO - Initialisation

- As with a GA, the initial set of points can be determined randomly or via more sophisticated sampling strategies.

- The velocities are also randomly initialized, generally using some fraction of the domain size.

- Particles must be prevented from going beyond constraint bounds. If a particle reaches a boundary and has a velocity pointing out of bounds, reset to velocity to zero or reorient it toward the interior for the next iteration.

- It is also helpful to impose a maximum velocity. If the velocity is too large, the updated positions are unrelated to their previous positions, and the search is more random.

- The maximum velocity might also decrease across iterations to shift from exploration toward exploitation.

PSO - Convergence

- Distance between each particle and the best particle

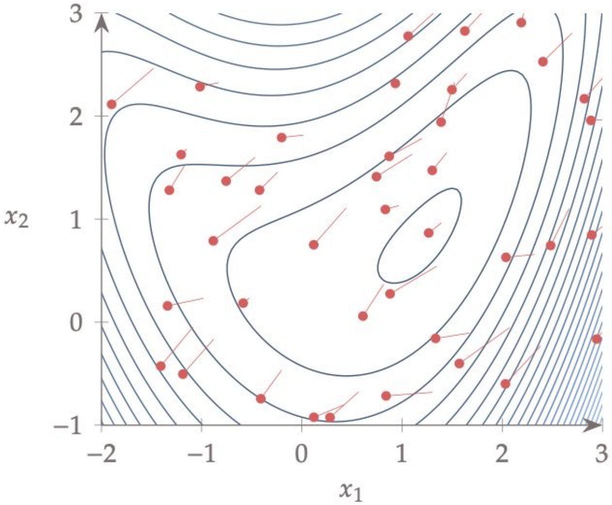
- Best particle's objective function value changes for the last few iterations

- Difference between the best and worst particle.

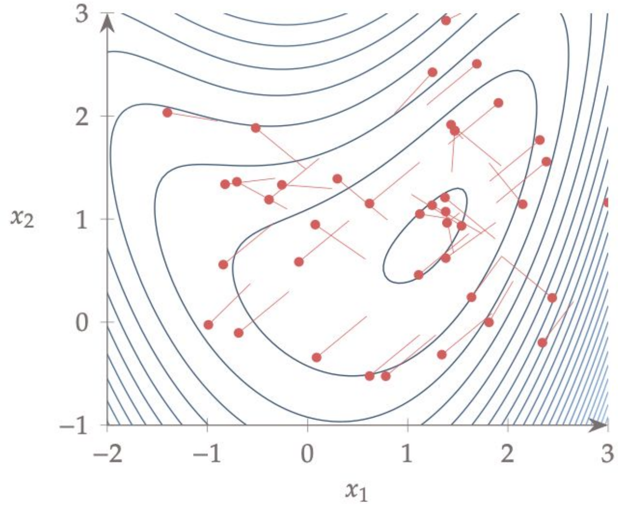
- Another alternative is to check whether the velocities for all particles are below some tolerance.

- Some of these criteria that assume all the particles congregate (distance, velocities) do not work well for multimodal problems. In those cases, tracking only the best particle's objective function value may be more appropriate.

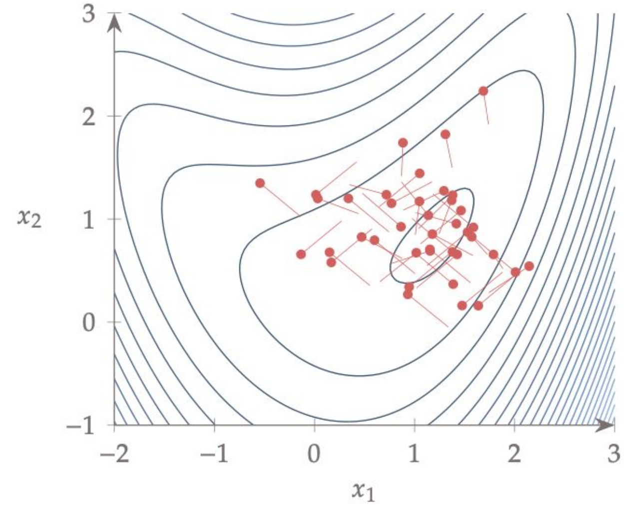
PSO - Results



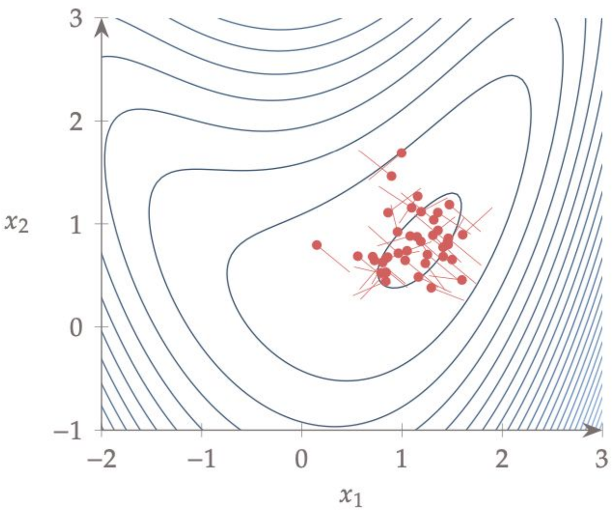
$k = 0$



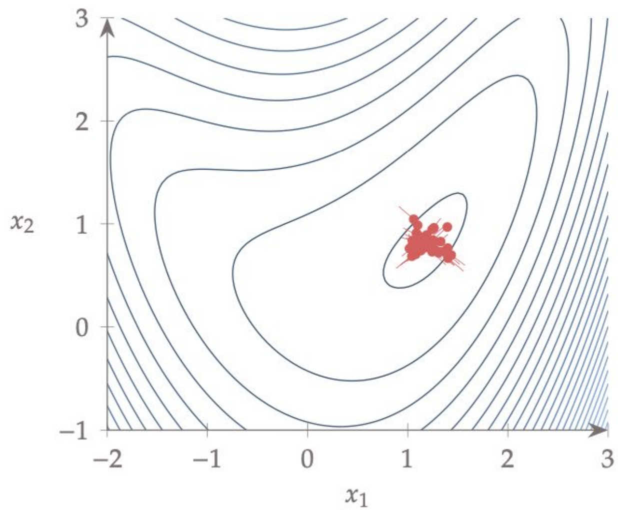
$k = 1$



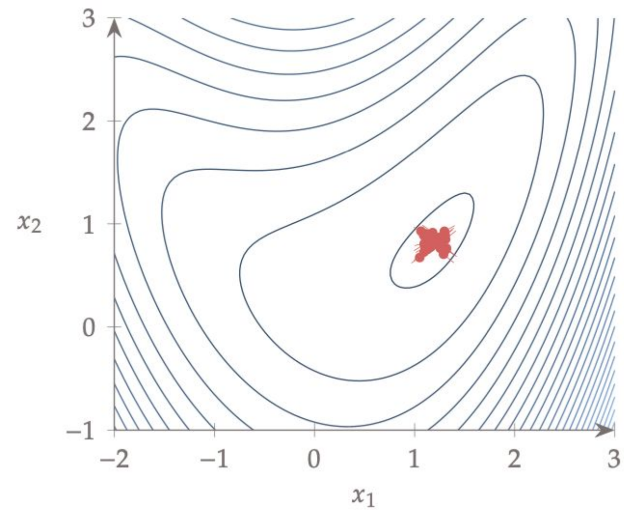
$k = 3$



$k = 5$



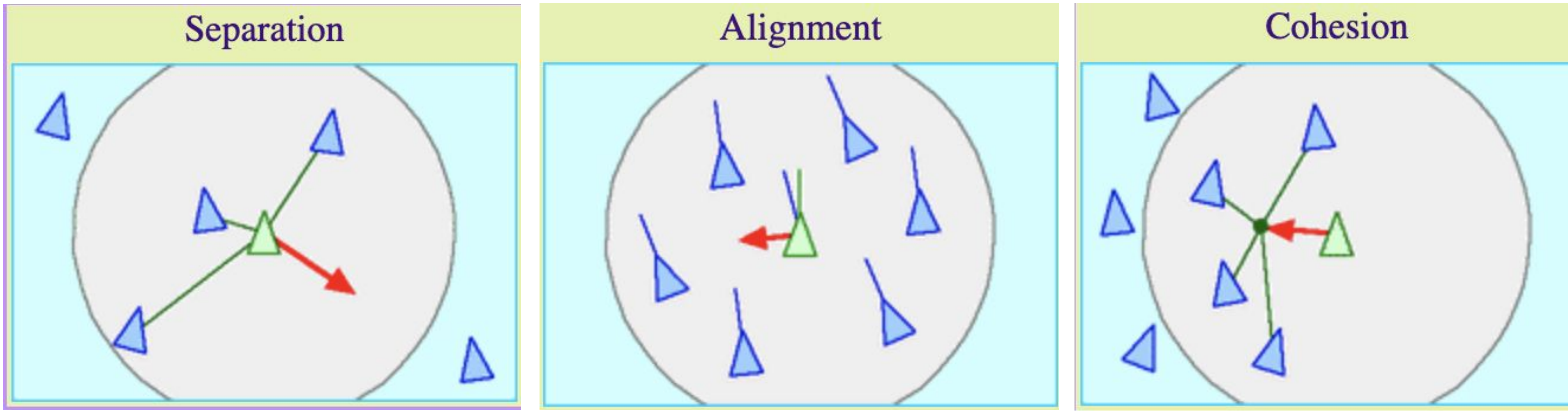
$k = 12$

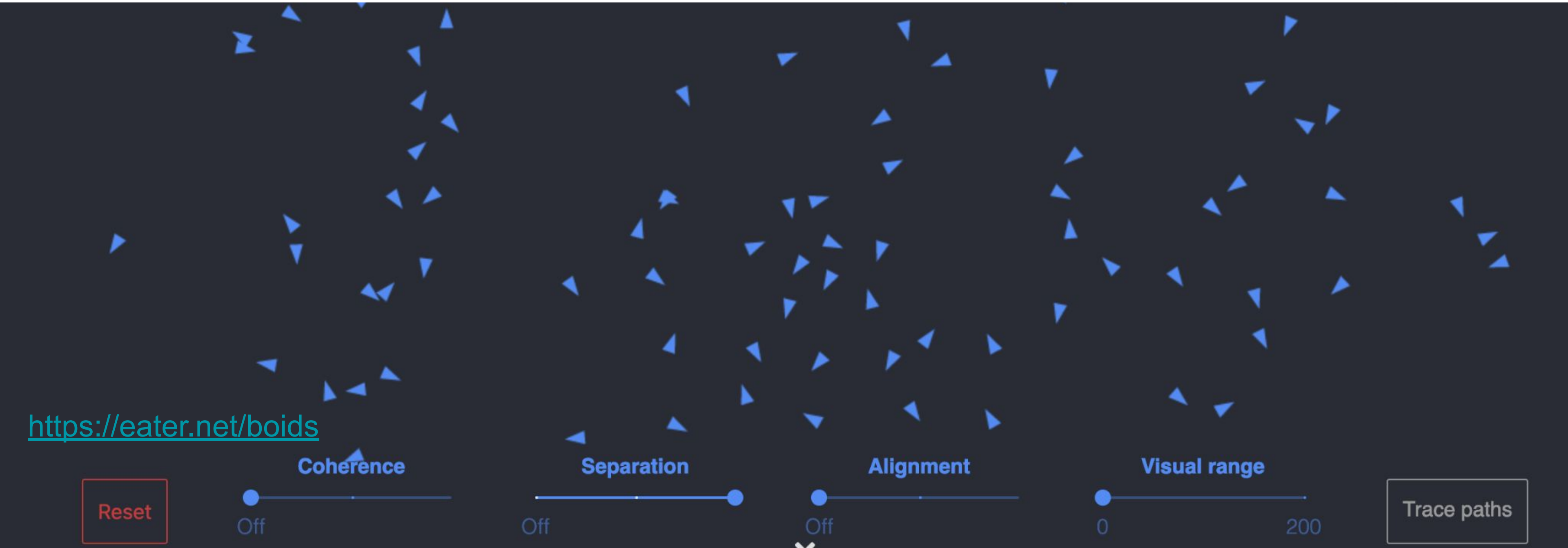


$k = 17$

Boids

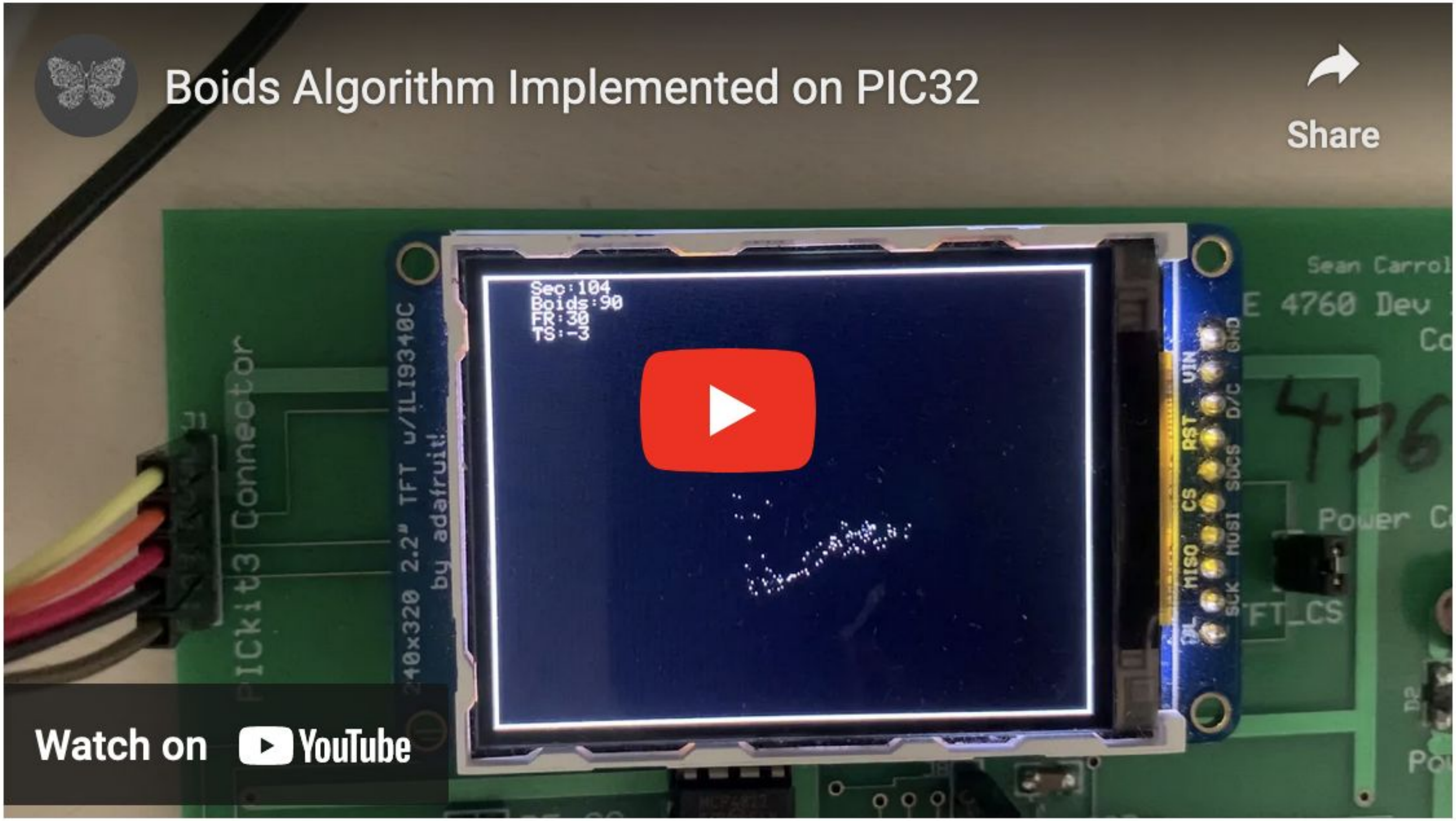
https://cs.stanford.edu/people/eroberts/courses/soco/projects/2008-09/modeling-natural-systems/boids.html





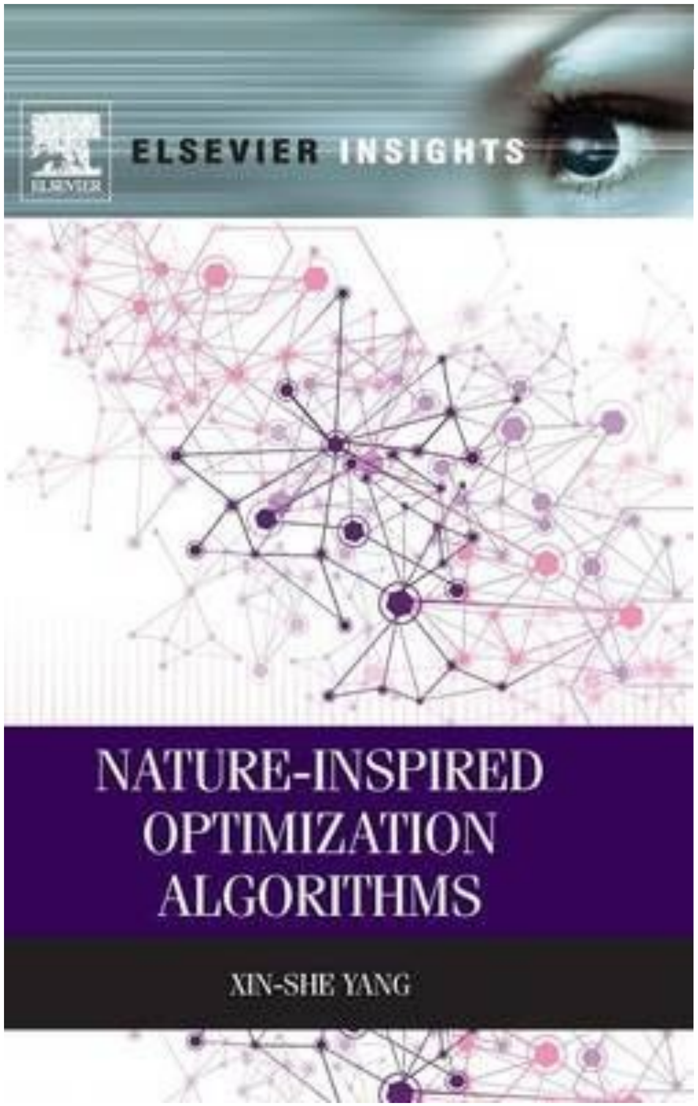
https://eater.net/boids

Boids on a PIC



https://people.ece.cornell.edu/land/courses/ece4760/labs/s2021/Boids/Boids.html

Other Nature Inspired Algorithms



Other Nature Inspired Algorithms

www.nature.com/scientificreports

scientific reports

Check for updates

OPEN

Hippopotamus optimization algorithm: a novel nature-inspired optimization algorithm

https://www.nature.com/articles/s41598-024-54910-3

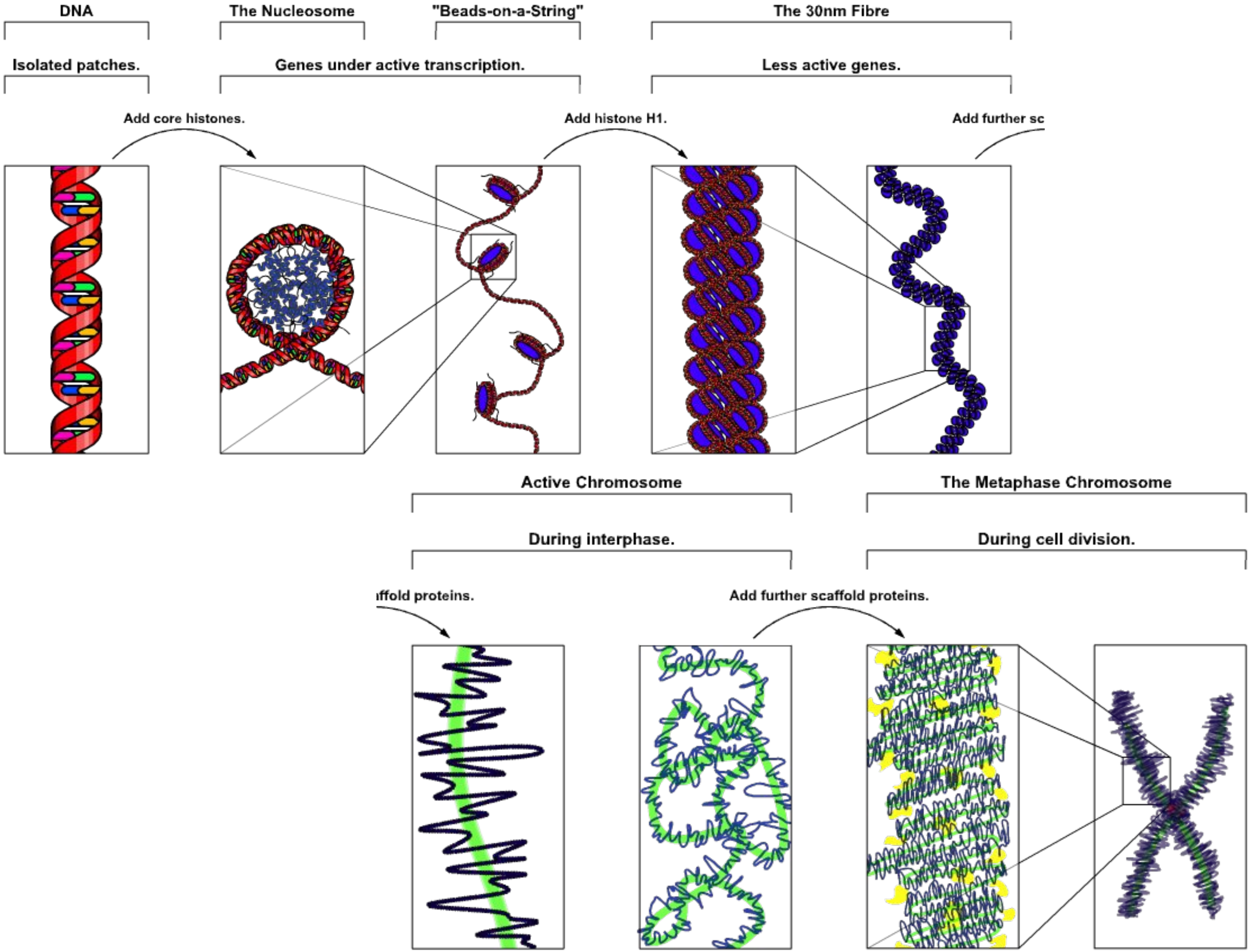
The “Real” Evolutionary Algorithm

- What constitutes an evolutionary algorithm is not well defined.

- Evolutionary algorithms are inspired by processes that occur in nature or society. There is a plethora of evolutionary algorithms in the literature, thanks to the fertile imagination of the research community and a never-ending supply of phenomena for inspiration.

- These algorithms are more of an analogy of the phenomenon than an actual model. They are, at best, simplified models and, at worst, barely connected to the phenomenon.

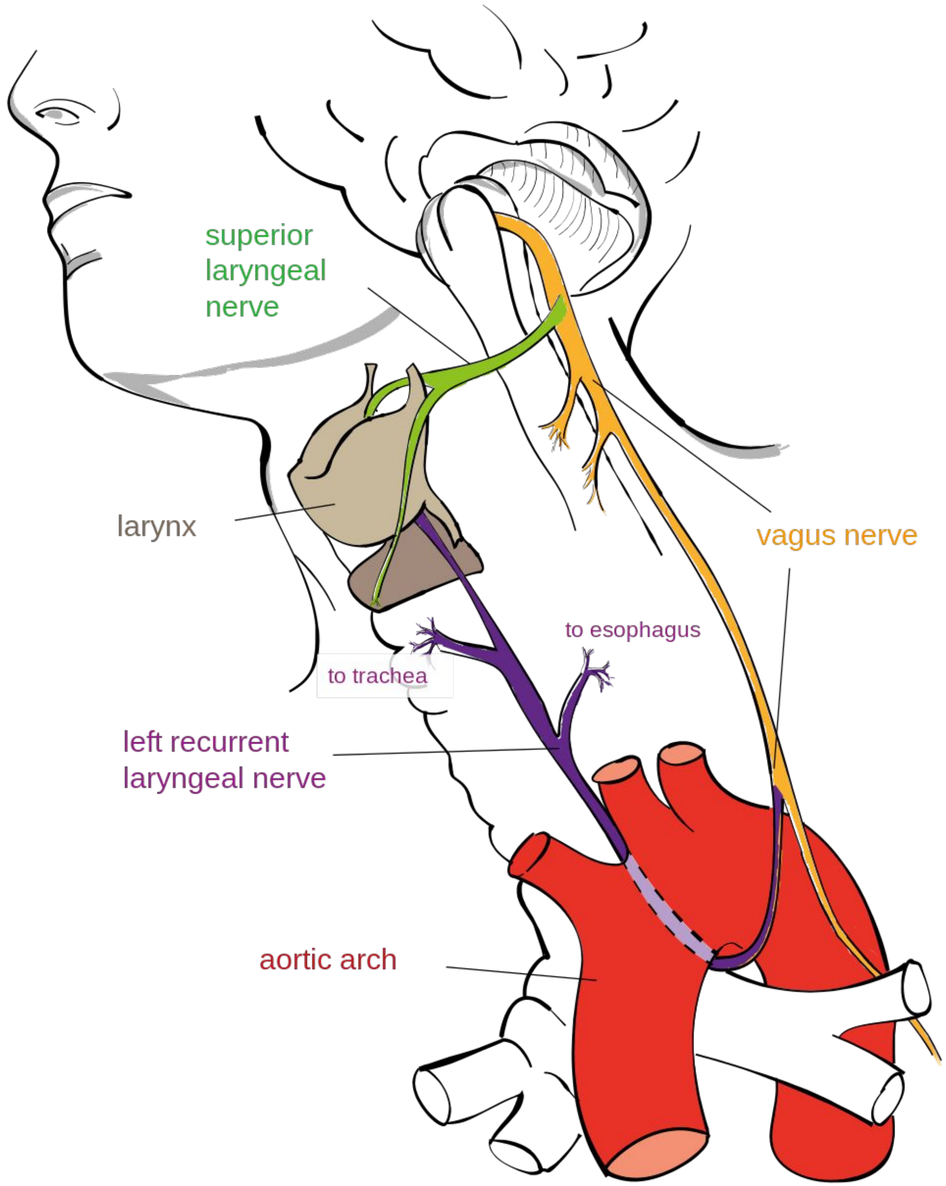
- Nature-inspired algorithms tend to invent a specific terminology for the mathematical terms in the optimisation problem.



DNA
The Nucleosome
"Beads-on-a-String"
The 30nm Fibre
Isolated patches.
Genes under active transcription.
Less active genes.
Add core histones.
Add histone H1.
Add further sc
Active Chromosome
The Metaphase Chromosome
During interphase.
During cell division.
affold proteins.
Add further scaffold proteins.

Recurrent laryngeal nerve

https://en.wikipedia.org/wiki/Recurrent_laryngeal_nerve



Exercises

- Come up with your own crossover strategies.
- Think of other ways to handle constraints in a GA besides the ones described in the lecture slides.
- Pick some points in 2D, and parameter values for the PSO algorithm and sketch the next point that the algorithm takes you to.

Thanks!

Mutant Jeans